Vinicius Vaz Cascais

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Professional Summary

Software Engineer working on **cloud-native infrastructure**. At Capgemini Engineering, with Nokia as the client, I perform **Helm rolling upgrades on Kubernetes** for production telecom systems **serving 5M+ subscribers** and configure traffic routing for highly available services. Systems foundation from 42 Lisboa (C/C++, Linux, processes) and experience with Docker and Nginx.

Professional Experience

Software Engineer - Capgemini Engineering *July 2026 - Present | Lisbon*

- Executed **Helm rolling upgrades** on **Kubernetes** for 5G/4G subscriber registers (AUSF/UDM/HSS/HLR) in a network **serving 5M+ subscribers**, running ~2 production rollouts per month with a zero-downtime design
- Troubleshot deployment issues during live upgrades using kubectl and logs
- Traffic routing using **Kubernetes Ingress** and internal networking to support **high availability** across sites
- Used internal **LLM tools** in the incident troubleshooting workflow to speed up diagnosis by ~3 hours

Full Stack Developer (Intern) - Ebinar *July 2025 - October 2025 | Lisbon*

- Built an Excel-to-MySQL import tool replacing manual imports, with filterable logs for users; **reduced manual import effort by ~25%** and cut data inconsistencies.

Education

42 Lisboa - Lisbon *Present*

Software Engineering Programme

Focus on C/C++, Object-Oriented Programming (OOP), system fundamentals and low-level memory management

IEFP - Lisbon *July 2025*

Network and Systems Administration – Technical Specialist Course

Built and maintained Windows and Linux servers and configured networks with Cisco Packet Tracer

Technical Skills

Languages: C, C++, Bash, PHP, SQL, GO

Cloud & Containers: Kubernetes, RedHat OpenShift, Docker, Helm, Terraform, AWS, Azure, Google Cloud, VMware

Systems & Infrastructure: Linux (RHEL, Arch, Ubuntu, Debian), Server Administration, Nginx, CI/CD Pipelines

A.I & Developer Tools: Claude Code, Cursor, Ollama (local LLMs), ChatGPT, Perplexity, Machine Learning

Networking: TCP/IP, DNS, HTTP, Kubernetes Ingress, Cisco Packet Tracer

Databases: MySQL/MariaDB

Tools: Git, GDB, Valgrind, Claude Code, Cursor, Ollama (local LLMs)

Projects

Photo Firefly - Cloud Based Photography Website

- Built and deployed a **full-stack** photography platform on **AWS**, with all infrastructure written in **Terraform** (EC2, EBS, S3, ECR, Route 53, IAM)
- Designed a **low-cost single-instance architecture** on AWS, matching SQLite's single-writer constraint with one small instance and persistent EBS; documented a scale-out path to **ECS Fargate, ALB and CloudFront**
- Secured the Photo Firefly AWS deployment with a **least-privilege, SSH-free design**: Secrets Manager and SSM for config, **IAM instance roles** replacing static keys, and a **private encrypted S3 bucket** for media, reducing security risks.
- Integrated **Stripe Checkout** with signed webhooks, **Google OAuth/Calendar APIs**, and SMTP email. Wrote a Vitest suite covering payments, file access and availability, plus tests that keep secrets out of client-side code

Inception – Docker Compose Infrastructure

- Designed and deployed a multi-container stack with Docker &Docker Compose, Nginx, MariaDB and WordPress.

Webserver – HTTP/1.1 Server in C++

- Built an HTTP/1.1 server from scratch with no external libraries, using a single non-blocking **poll()** event loop serving multiple ports, static files, uploads and CGI (Python, Node.js), handling malformed and slow clients.

Minishell – Custom Unix Shell in C

- Built a Unix-like shell in C with processes, pipes, redirections, environment variables and built-ins; added logical operators (&&, ||) and wildcard expansion.

Languages

Portuguese (Mother Tongue), English (C2), French (A2)

Certifications

Nokia Kubernetes certification (Level 3)

Nokia CNF Openshift (L1) Azure Foundations (L1)